VoltGuard: Topology-Aware Voltage-Risk Screening with Conformal Calibration for Active Distribution Networks

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Abstract

High photovoltaic penetration, electric-vehicle charging, and flexible demand make voltage-risk screening in active distribution networks increasingly uncertain. VoltGuard provides a calibrated screening layer that combines a squared-voltage LinDistFlow physical backbone, topology-aware residual learning, and topology/PV/loading conditioned split conformal calibration. The method converts pre-dispatch operating forecasts into voltage-risk intervals that are sharp enough for scenario triage while maintaining high recall of voltage-limit crossings. Experiments on IEEE 33-bus and IEEE 69-bus distribution feeders, with an expanded IEEE 118-bus stress audit, show that topology/PV/loading conditioned calibration improves the coverage-sharpness-risk tradeoff relative to global conformal calibration, tree-based point screens, and a neural graph residual ablation. On the representative 33/69 random split, VoltGuard achieves 0.00011 p.u. RMSE, 93.66% empirical coverage for nominal 90% intervals, 99.89% bus-level violation recall, and one missed bus-level violation. Across three random seeds, the method reduces mean missed violations from 1.67 to 0.33 while cutting mean interval width from 0.00127 to 0.00049 p.u. Runtime measurements show 0.254 ms/scenario online screening versus 1725 ms/scenario for the AC-audited grid-search backend, making VoltGuard a practical front end for prioritizing downstream AC analysis.

Keywords

Physics-informed machine learning; Conformal prediction; Topology-aware learning; Active distribution networks; Voltage security; Risk screening; Renewable hosting

1. Introduction

Active distribution networks are increasingly operated close to voltage limits while being asked to host more renewable generation. Rooftop PV, electric-vehicle charging, flexible loads, storage, and feeder switching create operating points that vary faster than classical planning margins were designed to handle. Voltage violations are costly in both directions: undervoltage degrades service quality and equipment operation, whereas overvoltage can force renewable curtailment or protection actions. The resulting problem is not only a voltage-estimation problem: operators need calibrated uncertainty around voltage-risk estimates before deciding which scenarios deserve downstream AC analysis.

Machine-learning voltage predictors are attractive because they can rapidly map forecasts and pseudo-measurements to voltage estimates. However, point accuracy is not enough for operation. A predictor with low RMSE can still miss rare voltage-limit crossings, and a globally conservative interval can produce so many false alarms that the operator no longer trusts it. The useful operational object is therefore a calibrated voltage-risk interval: an interval that is sharp enough to screen many safe scenarios, yet conservative enough to reduce missed violations near the 0.95-1.05 p.u. operating band.

This paper frames the task as calibrated voltage-risk screening. The proposed method, VoltGuard, uses a LinDistFlow physical backbone to encode feeder structure, learns a topology-aware residual correction from available operating features, and then uses split conformal calibration to convert residual errors into voltage intervals. The contribution is positive and operational: a fast, auditable screen that ranks scenarios and buses before AC analysis is invoked.

The contribution is intentionally not "a stronger GNN." A lightweight neural graph residual is included as an ablation, but the selected estimator is the topology-aware residual learner with conditioned conformal calibration because it gives the best risk-sharpness tradeoff in the executed experiments.

The paper makes three contributions:

1. A reproducible physics-informed topology-aware residual model that corrects a squared-voltage LinDistFlow backbone using only operating information available before AC labels are observed.
2. A topology/PV/loading conditioned split conformal calibration procedure with shrinkage fallback for voltage-risk intervals.
3. A focused screening evaluation on IEEE 33-bus and IEEE 69-bus distribution feeders showing how calibrated intervals affect coverage, interval width, false alarms, and missed-voltage-risk behavior.

The remaining audits, including EV-conditioning sensitivity, calibration-budget checks, candidate-action pruning, and high-PV hosting stress, are supporting application extensions. They are not used to define the central method claim.

1.1. Scope and Companion-Paper Positioning

This manuscript is the core screening-method paper. A companion DMS integration paper specifies deployment contracts, queue semantics, fallback rules, and operator audit records for using calibrated voltage-risk screens in control centers. A separate LinDistFlow-global-quantile note defines a minimal reference baseline. The three papers are parallel companion studies: this paper answers how to estimate calibrated voltage-risk intervals, the DMS paper answers how such intervals should be embedded in operations, and the baseline note answers what a transparent physical proxy plus one pooled quantile can achieve.

VoltGuard gives calibrated voltage-risk intervals under a stated feeder/scenario/split/calibration protocol. It complements AC power flow, OPF, and MPC by deciding which scenarios deserve immediate downstream analysis. AC feasibility, final dispatch, protection action, and closed-loop control remain the responsibility of those downstream tools.

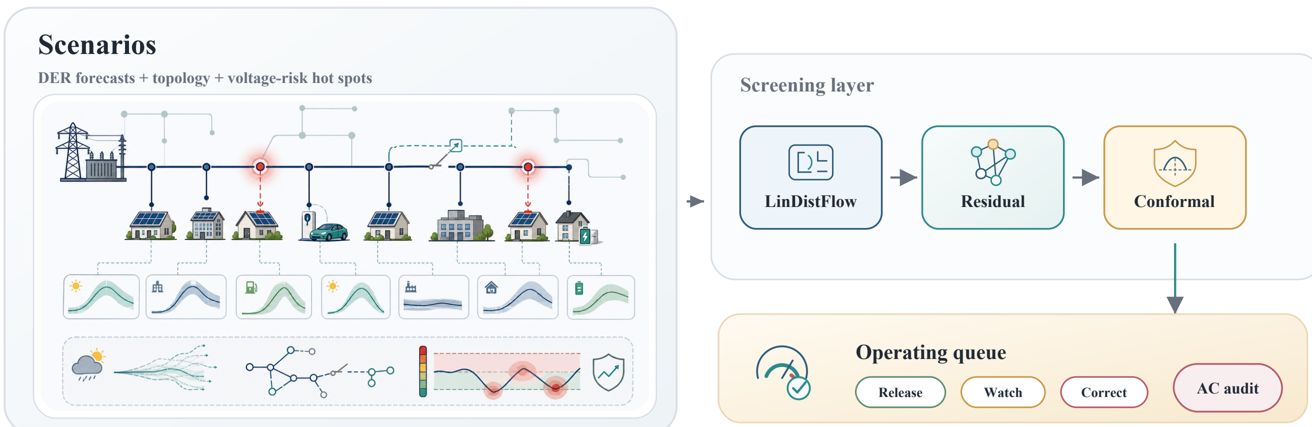

Figure 1: Graphical abstract and VoltGuard screening workflow. The figure summarizes the active-feeder inputs, LinDistFlow physical backbone, topology-aware residual correction, conformal voltage-risk intervals, and downstream AC-audited corrective optimization interface.

2. Related Work

Classical distribution-voltage assessment uses AC power flow, distribution state estimation, optimal power flow, Volt-VAR control, and conservative operating margins. Distribution feeder reconfiguration and DistFlow-style approximations trace back to radial-feeder power-flow models used for loss reduction and load balancing [1]. LinDistFlow and related approximations are valuable because they expose how downstream active and reactive power flows drive voltage drops. Their accuracy degrades under losses, reverse power flow, unmodeled imbalance, and uncertain injections, but they remain useful physical backbones for fast screening. OPF and distributed control remain the stronger corrective layers when an operating point requires actual dispatch decisions [2,3].

Physics-informed learning for power systems embeds network equations, line parameters, topology, or power-flow residuals into data-driven predictors. This improves interpretability and can reduce the burden on a black-box model. Recent physics-informed graph learning work for power-system state estimation also illustrates the value of combining electrical structure with learnable corrections [4]. In this work, the residual target is deliberately smaller than direct voltage prediction: the learner corrects a physical voltage proxy rather than replacing feeder physics.

Graph neural networks are natural for power grids because buses and lines form a graph. General graph-convolution and inductive representation-learning methods provide the modeling background for message passing on networks [5,6]. Many grid-learning studies emphasize point prediction, state estimation, or OPF acceleration. This paper treats graph-neural modeling as relevant background but not as the main claim. The executed neural graph residual is retained as an ablation and is not used to justify the title or contribution.

Conformal prediction converts arbitrary predictors into calibrated prediction sets or intervals under exchangeability assumptions [7,8]. Conformalized quantile regression further shows why interval efficiency matters, not only nominal coverage [9]. Recent conformalized GNN work highlights two difficulties that matter here: graph-structured samples may not be exchangeable in the usual i.i.d. sense, and intervals can become too wide if calibration ignores topology or operating regime [10,11]. VoltGuard addresses this operationally by conditioning calibration on feeder, PV, and loading families, then reporting per-family coverage and interval width rather than hiding all calibration behavior in a single global number.

Table 1 positions the paper relative to representative technical lines. The comparison is not a claim that every cited method solves the same screening task; it identifies the missing combination that motivates VoltGuard.

Table 1: Positioning relative to related work.

Line of work	Physics model	Topology-aware learning	Calibrated interval	Screening/DMS interface
AC OPF and distribution control [2,3,14-16]	yes	model based	no	corrective backend
Learning-assisted OPF/control [17,18,20]	partial	sometimes	usually no	dispatch-oriented
Physics-aware state estimation [4,19]	yes	yes	usually no	estimation-oriented
Conformal prediction and CQR [7-9,22-24]	model agnostic	no	yes	task agnostic
Conformalized GNNs [10,11]	no explicit feeder physics	yes	yes	graph prediction
VoltGuard	LinDistFlow backbone	yes	T/PV/L conditioned	voltage-risk screening

3. Problem Formulation

Consider a distribution feeder

$$\mathcal{G} = (\mathcal{N}, \mathcal{E}),$$

where $\mathcal{N}$ is the bus set and $\mathcal{E}$ is the branch set. A root or slack bus 0 defines branch orientation: each branch $(i, j) \in \mathcal{E}$ is directed from upstream bus i toward downstream bus j. Each branch has resistance r_{ij} and reactance x_{ij}. Let $v_{i,t}$ be voltage magnitude at bus i in scenario t, and let

$$u_{i,t} = v_{i,t}^2$$

be squared voltage magnitude. The slack voltage is fixed to $v_{0,t} = 1.0$ p.u. in the executed simulations, hence $u_{0,t} = 1.0$.

The operating features available before screening are load forecasts or pseudo-measurements $p_{i,t}^d, q_{i,t}^d$, PV forecasts $p_{i,t}^g$, EV-load indicators, feeder topology, and local/neighbor electrical features. Net injection is written as

$$p_{i,t} = p_{i,t}^d - p_{i,t}^g, \qquad q_{i,t} = q_{i,t}^d - q_{i,t}^g.$$

Positive $p_{i,t}$ denotes net demand. AC power-flow labels $v_{i,t}^\star$ are used only for supervised training, calibration scoring, and test evaluation. Test labels are never used to construct features or conformal radii.

The goal is to produce calibrated voltage intervals

$$\hat{\mathcal{J}}_{i,t}^{1-\alpha} = [\hat{v}_{i,t}^L, \hat{v}_{i,t}^U]$$

that are sharp enough for operation while limiting missed voltage violations. The bus-level risk flag is

$$\rho_{i,t} = \mathbf{1}\{\hat{v}_{i,t}^L < v^{\min} \text{ or } \hat{v}_{i,t}^U > v^{\max}\},$$

with $v^{\min} = 0.95$ p.u. and $v^{\max} = 1.05$ p.u. The scenario-level flag is

$$\rho_t^{\max} = \max_{i \in \mathcal{N}} \rho_{i,t}.$$

For prioritization, VoltGuard also computes a scenario-level interval risk score

$$\eta_t = \max_{i \in \mathcal{N}} \left[\max(0, v^{\min} - \hat{v}_{i,t}^L), \max(0, \hat{v}_{i,t}^U - v^{\max})\right].$$

Large η_t means that at least one calibrated bus interval crosses a voltage limit by a larger margin, so the scenario should be considered earlier by downstream AC-audited corrective optimization.

4. Methodology

4.1. LinDistFlow Backbone

For each scenario, approximate downstream branch flows are computed from the forecast net injections and the oriented radial feeder tree:

$$\hat{P}_{ij,t} = \sum_{k \in \mathcal{D}(j)} p_{k,t}, \qquad \hat{Q}_{ij,t} = \sum_{k \in \mathcal{D}(j)} q_{k,t},$$

where $\mathcal{D}(j)$ is the downstream subtree rooted at bus j. The squared-voltage LinDistFlow recursion is

$$\hat{u}^{lin}_{j,t} = \hat{u}^{lin}_{i,t} - 2(r_{ij}\hat{P}_{ij,t} + x_{ij}\hat{Q}_{ij,t}), \qquad \hat{u}^{lin}_{0,t} = 1.$$

The voltage-magnitude proxy is

$$\hat{v}^{lin}_{i,t} = \sqrt{\max(\hat{u}^{lin}_{i,t}, \epsilon)},$$

where $\epsilon > 0$ prevents numerical issues. The executable prototype computes this squared-voltage backbone explicitly and stores both $\hat{u}^{lin}_{i,t}$ and $\hat{v}^{lin}_{i,t}$. These quantities are used as residual targets and model inputs together with bus position, voltage base, degree, load, PV, net demand, neighbor load/PV/net demand, load scale, PV penetration, and EV scale. No test AC voltage label is used to build the LinDistFlow proxy, conformal radius, or test-time risk flag.

4.2. Topology-Aware Residual Learner

VoltGuard predicts a voltage-magnitude residual

$$\Delta\hat{v}_{i,t} = f_\theta(x_{i,t}, z_{i,t}, g_i),$$

where $x_{i,t}$ contains local bus operating features, $z_{i,t}$ contains neighbor-aggregated electrical features, and g_i contains topology features such as normalized bus index, slack indicator, voltage base, and degree. The final point estimate is

$$\hat{v}_{i,t} = \hat{v}^{lin}_{i,t} + \Delta\hat{v}_{i,t}.$$

In the executable implementation, f_θ is an ExtraTrees residual learner trained on $v^\star_{i,t} - \hat{v}^{lin}_{i,t}$. This choice is deliberate: it is stable on the executed feeder/scenario families and outperforms the lightweight neural graph residual on the point-error, interval-width, and false-alarm metrics used for the selected model.

4.3. Physics-Informed Objective

The supervised residual objective is

$$\mathcal{L}_{sup} = \frac{1}{|\mathcal{T}||\mathcal{N}|} \sum_{t\in\mathcal{T}} \sum_{i\in\mathcal{N}} \left(\hat{v}_{i,t} - v^{\star}_{i,t}\right)^2 .$$

The physics residual used to define the reproducible model class is a branch-level voltage-drop mismatch in squared-voltage space:

$$\mathcal{R}_{drop} = \frac{1}{|\mathcal{T}||\mathcal{E}|} \sum_{t\in\mathcal{T}} \sum_{(i,j)\in\mathcal{E}} \left(\hat{u}_{j,t} - \hat{u}_{i,t} + 2(r_{ij}\hat{P}_{ij,t} + x_{ij}\hat{Q}_{ij,t})\right)^2 .$$

Here $\hat{u}_{i,t} = \hat{v}^2_{i,t}$ after residual correction, and $\hat{P}, \hat{Q}$ are computed from forecast injections and feeder orientation, not from test labels. A smoothness penalty may be added as

$$\mathcal{R}_{smooth} = \frac{1}{|\mathcal{T}||\mathcal{E}|} \sum_{t,(i,j)} \frac{(\hat{v}_{i,t} - \hat{v}_{j,t})^2}{r^2_{ij} + x^2_{ij} + \epsilon}.$$

The full conceptual objective is

$$\mathcal{L} = \mathcal{L}_{sup} + \lambda_{drop}\mathcal{R}_{drop} + \lambda_{smooth}\mathcal{R}_{smooth}.$$

The current tree-based residual implementation realizes the same philosophy through physics-derived inputs and residual targets rather than through gradient-based neural optimization of all penalty terms. The physics residual is still reported explicitly so that a neural or differentiable implementation can reproduce the same objective without changing the paper's model boundary.

4.4. Split Conformal Calibration

For a calibration set $\mathcal{C}$ disjoint from training and testing, the nonconformity score is

$$s_{i,t} = |v^{\star}_{i,t} - \hat{v}_{i,t}|.$$

For nominal miscoverage α, global split conformal calibration uses

$$\hat{q}_{1-\alpha} = \text{Quantile}_{\lceil(|\mathcal{C}|+1)(1-\alpha)\rceil/|\mathcal{C}|}\{s_{i,t} : (i,t) \in \mathcal{C}\}.$$

The global interval is

$$\hat{\mathcal{I}}^{1-\alpha}_{i,t} = [\hat{v}_{i,t} - \hat{q}_{1-\alpha}, \hat{v}_{i,t} + \hat{q}_{1-\alpha}].$$

The finite-sample conformal statement relies on exchangeability between calibration and test samples [7,8]. In this paper, that means the stated feeder/scenario family, split rule, and calibration protocol. The paper does not claim unconditional safety under arbitrary topology switching, time drift, PV distribution shift, or field deployment.

4.5. *Topology/PV/Loading Conditioned Calibration*

Voltage residual distributions vary by feeder, PV level, and loading regime. VoltGuard therefore partitions calibration samples into families

$$g(i,t) = \{\text{feeder}, \text{PV bin}, \text{load bin}\}.$$

For each family g, a family quantile $\hat{q}_{g,1-\alpha}$ is computed. To avoid overconfident intervals in small families, VoltGuard uses shrinkage toward the global quantile:

$$\tilde{q}_{g,1-\alpha} = \omega_g \hat{q}_{g,1-\alpha} + (1-\omega_g)\hat{q}_{1-\alpha}, \qquad \omega_g = \frac{n_g}{n_g + n_0},$$

where n_g is the number of calibration samples in family g and n_0 is the minimum-group scale. The experiments compare global, PV-conditioned, topology-conditioned, topology+PV+loading conditioned, and no-shrinkage variants. EV scale is included in the point-estimator feature set, but it is not automatically used as a conformal grouping key because it can fragment family sample sizes. The EV-conditioned ablation in the results tests this choice directly.

4.6. *Two-Stage Operating Interface*

VoltGuard is a screening layer. It does not select final dispatch as an OPF or MPC controller. In the two-stage interface:

1. VoltGuard produces calibrated bus-level intervals and scenario-level risk flags.
2. Risky scenarios are passed to an AC-audited corrective layer, such as discrete grid search, OPF, or MPC.
3. When the downstream layer has limited compute budget, scenarios are ranked by η_t so that larger interval-limit crossings are audited first.

The simple risk-trigger action used for diagnostic operating value applies 10% flexible-load relief when undervoltage risk appears and 10% PV curtailment when overvoltage risk appears. A stronger AC-audited grid search is then reported as a downstream corrective benchmark.

5. Experimental Design

The main experiments use IEEE 33-bus and IEEE 69-bus distribution feeders. IEEE 118-bus is retained as a supplementary stress test because it is not a canonical distribution feeder. For each feeder, 300 AC-solvable scenarios are generated with randomized load scale, PV penetration, EV demand, and PV siting. AC power flow provides reference labels.

Four split protocols are executed:

- Random interpolation: scenario-disjoint 60/20/20 train/calibration/test split.
- Synthetic time-block: scenario IDs are treated as ordered blocks.
- PV-penetration shift: low/medium PV scenarios train, medium-high PV calibrates, and high PV tests.

- Topology-held-out transfer: IEEE 33-bus trains the residual learner, while IEEE 69-bus calibration and testing evaluate feeder transfer.

The main random split is repeated with seeds 7, 17, and 42. The representative run uses seed 7, and all raw predictions, conformal scores, family labels, and risk flags are saved for that run.

Baselines include the squared-voltage LinDistFlow physical backbone, random forest, gradient boosting, boosting plus global conformal calibration, VoltGuard conformal variants, and a neural graph residual ablation. These baselines cover three screening families requested by reviewers: transparent physical screening, standard tree-based machine learning, and graph-neural residual learning. Metrics are bus-level MAE/RMSE, coverage, interval width, violation precision/recall/F1, false-alarm rate, missed violations, scenario-level recall/false alarms, and family-level calibration behavior. Downstream operating audits remain project artifacts outside the main method submission.

For readability, result tables use a fixed compact label set: LDF denotes the LinDistFlow physical backbone, RF denotes random forest, GB denotes gradient boosting, Boost-GC denotes boosting with global conformal calibration, GB-quantile denotes gradient-boosted quantile regression, GP-UQ denotes the Gaussian-process uncertainty baseline, T/PV/L denotes topology/PV/loading conditioning, full T/E denotes the full topology/electrical feature family, and GNN ablation denotes the neural graph residual diagnostic.

5.1. Comparison with State-of-the-Art Screening Baselines

The experiments include six competing or diagnostic approaches. LinDistFlow is the direct physical proxy and represents a voltage-sensitivity-style screen without learning. Random forest and gradient boosting are standard tabular ML screens using the same pre-dispatch features. Boosting plus global conformal calibration is a conformalized point-prediction baseline. The neural graph residual ablation tests whether message passing is competitive under the same data budget. VoltGuard is the proposed topology-aware residual learner with topology/PV/loading conditioned conformal calibration. The comparison table in Section 6 reports the same error, coverage, width, false-negative, and false-alarm quantities for these methods.

6. Results

6.1. Representative Random-Split Results

On the representative IEEE 33/69 random split, VoltGuard with topology+PV+loading conditioned calibration achieves the strongest screening tradeoff among the selected main estimators.

Table 2: Representative screening metrics.

Method	Calib.	RMSE	Cov.	Wid.	Recall	FA	Missed
LDF	none	0.00140	–	–	0.91640	0.00000	77
RF	none	0.00023	–	–	0.99023	0.00019	9
GB	none	0.00041	–	–	0.97286	0.00000	25
Boost-GC	Global	0.00041	0.90850	0.00110	0.99674	0.00250	3
VoltGuard	T/PV/L	0.00011	0.93660	0.00043	0.99891	0.00058	1

Table 2: Representative screening metrics.

Method	Calib.	RMSE	Cov.	Wid.	Recall	FA	Missed
GNN ablation	topology	0.00171	0.91340	0.00540	0.99783	0.02327	2

The neural ablation attains high recall but with substantially wider intervals, higher RMSE, and higher false-alarm rate. It is therefore kept as an ablation, not as the main estimator.

6.2. Branch-Level Physics Consistency Audit

Because VoltGuard corrects a LinDistFlow-style voltage proxy, we audit whether the saved voltage predictions remain consistent with the branch-level voltage-drop relation used in the methodology. For each representative test scenario, feeder branches are oriented from the slack bus, downstream forecast net injections are aggregated, and the residual

$$\hat{u}_j - \hat{u}_i + 2(r_{ij}\hat{P}_{ij} + x_{ij}\hat{Q}_{ij})$$

is computed on each branch. This is a LinDistFlow consistency audit, not an AC feasibility certificate; the AC-label row is not zero because the proxy ignores losses and other full AC effects.

Table 3: Physics-consistency audit.

Method	Variant	Scenarios	Drop RMSE	Drop MAE	95% max drop	Viol. drop RMSE
AC power-flow label	reference	120	0.000238	0.000101	0.002516	0.000272
Boost-GC	Global	120	0.000563	0.000379	0.004631	0.000621
VoltGuard	T/PV/L	120	0.000250	0.000122	0.002395	0.000279
GNN ablation	topology	120	0.003139	0.002437	0.014201	0.002904

VoltGuard substantially reduces the branch-drop RMSE relative to boosting plus global conformal and remains close to the AC-label audit residual, while the neural graph residual is substantially less consistent with the LinDistFlow drop relation. This supports the paper's physics-informed claim in the narrow sense used here: residual learning is anchored to a real LinDistFlow backbone, operating features, and topology in a way that improves calibrated screening without moving far from the physical voltage-drop structure.

6.3. Multi-Seed Random-Split Statistics

Across three random seeds, VoltGuard reduces missed violations relative to boosting plus global conformal while maintaining near-nominal coverage.

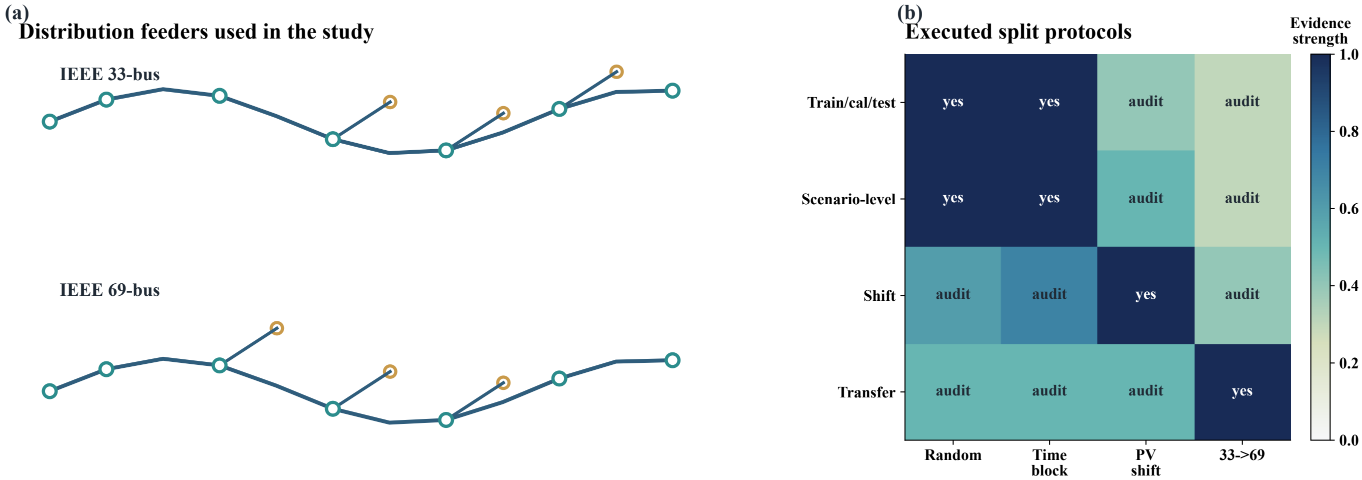

Figure 2: Main distribution feeders and generalization protocols. (a) IEEE 33-bus and 69-bus distribution feeders used as the main experimental systems; (b) executed split protocols for random interpolation, synthetic time-block, PV-shift, and topology-transfer audits.

Table 4: Multi-seed screening summary.

Method	Variant	RMSE mean	Coverage mean	Width mean	Recall mean	FA mean	Missed mean	Runs
Boost-GC	Global	0.00076	0.90468	0.00127	0.99811	0.00240	1.67	3
VoltGuard	T/PV/L	0.00017	0.92397	0.00049	0.99964	0.00171	0.33	3

The confidence intervals are reported in `multi_seed_summary.md`. The main interpretation is not that every seed dominates on every metric, but that conditioned calibration improves the risk-recall tradeoff without relying on a GNN claim.

We also report paired seed deltas relative to boosting plus global conformal, using the same split seed before averaging. Positive deltas mean VoltGuard is larger; negative deltas are better for RMSE, width, false alarms, and missed violations. On the random interpolation split, VoltGuard reduces missed bus-level violations by 1.33 on average, increases recall by 0.00152, and narrows intervals by 0.00078 p.u. On synthetic time-block splits, the paired missed-violation reduction is also 1.33. Under PV-shift, VoltGuard keeps recall slightly higher while coverage drops below nominal, reflecting the deliberately shifted test distribution. Under topology-held-out 33-to-69 transfer, missed violations are unchanged while intervals and false alarms are lower, but this is still a protocol-specific transfer result rather than a topology-invariant safety guarantee.

Table 5: Paired seed deltas relative to the global baseline.

Split	Runs	Delta coverage	Delta width	Delta recall	Delta FA	Delta missed
random interpolation	3	0.01928	-0.00078	0.00152	-0.00070	-1.33
synthetic time-block	3	0.04025	-0.00058	0.00140	-0.00110	-1.33
PV-penetration shift	3	-0.02652	-0.00044	0.00064	-0.00016	-0.33
topology-held-out 33-to-69	3	0.00242	-0.00078	0.00000	-0.00129	0.00

6.4. Feature/Residual Ablation

We add a feature/residual ablation to test whether the main estimator is actually topology-aware or merely a generic tree ensemble. All variants use the same topology/PV/loading conformal calibration and three random seeds. The local-only residual learner uses only bus-level operating quantities; the local+topology variant adds bus position, voltage base, degree, slack indicator, and feeder code; the full variant also includes neighbor load/PV/net-demand features. A direct full ExtraTrees model tests whether the residual formulation matters once the same full feature family is available.

Table 6: Residual feature-family ablation.

Variant	Feature family	Runs	RMSE	Cov.	Wid.	Recall	FA	Missed
direct	full T/E	3	0.00029	0.91111	0.00073	0.99848	0.00298	1.33
residual	local	3	0.00073	0.92195	0.00207	0.99621	0.00373	3.33
residual	local+T	3	0.00022	0.91694	0.00071	0.99964	0.00298	0.33
residual	full T/E	3	0.00017	0.92413	0.00050	0.99964	0.00171	0.33

The ablation supports the topology/electrical part of the claim strongly: removing topology and neighbor electrical features increases the mean missed bus-level violations from 0.33 to 3.33 and widens intervals from 0.00050 to 0.00207 p.u. The residual formulation should be read more narrowly. Relative to a direct full ExtraTrees predictor, the residual full model lowers RMSE (0.00017 versus 0.00029), interval width (0.00050 versus 0.00073), and false-alarm rate (0.00171 versus 0.00298), while both the full residual and local+topology residual have statistically comparable missed-violation counts. This is why the paper emphasizes physics-informed topology-aware residual screening rather than claiming that residualization alone dominates every operating metric.

6.5. Comparison with Competing Screening Methods

The reviewer-requested comparison table brings together the representative random-split results for physical, tree-based, conformal, statistical-UQ, and graph-residual screens. The quantile-regression and Gaussian-process rows are additional baselines run on the same seed-7 random split. The Gaussian-process baseline uses a 900-row training subsample for tractability and is included to test whether generic probabilistic regression gives useful screening intervals without feeder-specific residual structure.

Table 7: Comparison with competing screening methods.

Method	Interval	RMSE	Cov.	Wid.	Recall	FA	Missed
LDF	point only	0.00140	n/a	n/a	0.91640	0.00000	77
RF	point only	0.00023	n/a	n/a	0.99023	0.00019	9
GB	point only	0.00042	n/a	n/a	0.97286	0.00000	25
Boost-GC	pooled Q	0.00042	0.90850	0.00110	0.99674	0.00250	3
GB-quantile	5/95% Q	0.00035	0.83219	0.00444	0.99566	0.00135	4
GP-UQ	Gaussian	0.01165	0.97827	0.05108	1.00000	0.33353	0
GNN ablation	Topo-cond	0.00171	0.91340	0.00540	0.99783	0.02327	2
VoltGuard	T/PV/L	0.00011	0.93660	0.00043	0.99891	0.00058	1

The comparison clarifies the novelty boundary. Standard tree models can give competitive point accuracy, but their uncalibrated risk flags miss more violations. Global conformal calibration improves recall but is less sharp than conditioned calibration. Quantile regression gives intervals but under-covers in this split, while generic Gaussian-process intervals become too wide and produce many false alarms. The graph residual does not outperform the topology-aware residual learner. VoltGuard's advantage is therefore the combination of a physical voltage proxy, topology/electrical residual features, and conditioned conformal calibration rather than a generic ML model alone.

6.6. Conformal Ablation

The representative random split directly compares calibration variants.

Table 8: Conformal calibration variants.

Variant	Cov.	Wid.	Recall	FA	Missed
Global	0.94984	0.00053	0.99891	0.00115	1
PV-cond	0.94510	0.00049	0.99891	0.00058	1
Topo-cond	0.95523	0.00053	0.99891	0.00154	1
T/PV/L	0.93660	0.00043	0.99891	0.00058	1
No shrinkage	0.91258	0.00042	0.99891	0.00038	1

All VoltGuard conformal variants miss only one bus-level violation in this split. The topology+PV+loading conditioned variant is selected because it keeps that recall while producing the sharpest shrinkage-stabilized interval; the no-shrinkage version is marginally narrower but loses coverage, which is why shrinkage remains in the main method.

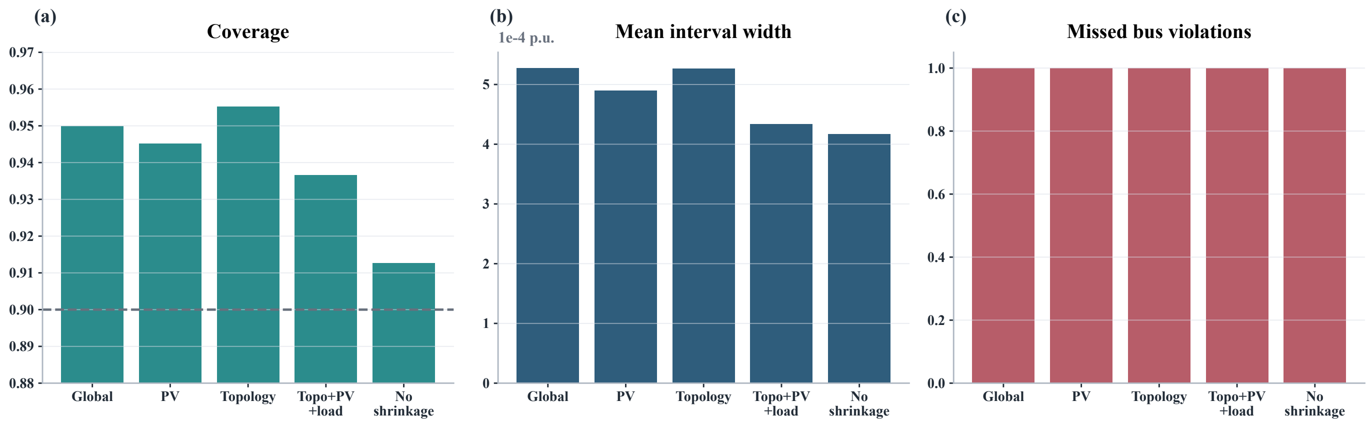

Figure 3: Conformal calibration variant comparison. (a) Empirical coverage across calibration variants; (b) mean interval width; (c) missed bus-level voltage violations.

6.7. Calibration-Budget Sensitivity

Conditioned conformal calibration is useful only if the calibration set covers the operating families that will appear at test time. We therefore add a scenario-level calibration-budget audit using the saved representative split. The residual model is not retrained. Instead, calibration scenarios are sampled without replacement at 10%, 25%, 50%, 75%, and 100% of the calibration split, and conformal radii are recomputed before evaluating the same held-out test set. Partial-budget cases are repeated 20 times. Sampling is done at the scenario level so buses from the same scenario are not treated as independent calibration draws.

Table 9: Calibration-budget sensitivity.

Method	Calib. frac.	Calib. scen.	Obs. fam.	Empty fam.	Cov.	Wid.	Recall	FA	Missed
Boost-GC	0.10	12	1.00	0.00	0.89410	0.00104	0.99457	0.00181	5.00
Boost-GC	1.00	120	1.00	0.00	0.90850	0.00110	0.99674	0.00250	3.00
VoltGuard	0.10	12	8.40	7.60	0.92903	0.00047	0.99902	0.00110	0.90
VoltGuard	0.25	30	13.35	2.65	0.93528	0.00044	0.99924	0.00091	0.70
VoltGuard	0.50	60	15.80	0.20	0.94022	0.00044	0.99919	0.00089	0.75
VoltGuard	0.75	90	16.00	0.00	0.94051	0.00044	0.99919	0.00078	0.75
VoltGuard	1.00	120	16.00	0.00	0.93660	0.00043	0.99891	0.00058	1.00

The budget audit supports deployment, but with a clear boundary. With only 12 calibration scenarios, the shrinkage/fallback construction prevents the screen from collapsing: coverage remains 0.92903 and the mean missed bus-level violations stay below one in this representative test set. However, the calibration set observes only 8.4 of the 16 test families on average, so many conditioned families are not actually calibrated locally. At 75% and full budget, all 16 families are observed. We therefore describe small-budget results as evidence for a robust fallback mechanism, not as evidence for family-wise coverage without family-level calibration diversity.

6.8. Asymmetric Conformal Audit

We also audit an asymmetric one-sided conformal variant because voltage limits are directional: undervoltage risk depends on the lower tail and overvoltage risk depends on the upper tail. This audit uses the same saved calibration and test predictions and replaces the symmetric absolute-residual radius with separate lower and upper one-sided conformal radii. It is not a new distribution-shift guarantee; it is a tail-aware calibration ablation under the same split and exchangeability assumptions as the main conformal analysis.

Table 10: Asymmetric conformal calibration audit.

Method	Calibration	Cov.	Wid.	Recall	FA	Missed
Boost-GC	symmetric	0.90850	0.001096	0.99674	0.00250	3
Boost-GC	asymmetric	0.90539	0.001162	0.99891	0.00327	1
VoltGuard Global	symmetric	0.94984	0.000527	0.99891	0.00115	1
VoltGuard Global	asymmetric	0.94624	0.000497	0.99891	0.00058	1
VoltGuard	symmetric	0.93660	0.000433	0.99891	0.00058	1
VoltGuard	asymmetric	0.91977	0.000411	1.00000	0.00058	0

For the main topology+PV+loading variant, the asymmetric audit narrows the mean interval by 0.000022 p.u. and removes the remaining missed bus-level violation while retaining scenario-level recall of 1.0. The cost is lower empirical coverage, 0.91977 rather than 0.93660, although still above the nominal 90% target in this representative split. We therefore keep symmetric conditioned conformal calibration as the main method and report the asymmetric variant as an optional tail-aware extension that may be useful when missed voltage violations are costlier than modest interval-efficiency loss.

6.9. EV-Conditioning Ablation

Because the scenarios also include EV charging perturbations, we add an EV-conditioned ablation to test whether EV scale should be added to the conformal family key. The result is negative for the current sample size. Adding EV to the topology+PV+loading family increases the number of families from 16 to 55, reduces the minimum calibration count from 198 to 33, and leaves seven empty test families. It does not reduce missed violations or improve recall; it increases interval width and false alarms. Thus EV demand remains a pre-dispatch feature for residual learning, but the conformal calibration key is kept at feeder/PV/loading granularity unless a larger EV-calibration set is available.

Table 11: EV-conditioning family ablation.

Conditioning key	Families	Min calib.	Empty fam.	Cov.	Wid.	Recall	FA	Missed
T/PV/L	16	198	0	0.93660	0.00043	0.99891	0.00058	1
T/PV/L+EV	55	33	7	0.94918	0.00048	0.99891	0.00154	1
T/PV/EV	31	33	1	0.94853	0.00050	0.99891	0.00173	1

Table 11: EV-conditioning family ablation.

Conditioning key	Families	Min calib.	Empty fam.	Cov.	Wid.	Recall	FA	Missed
PV/load/EV	32	33	0	0.94003	0.00045	0.99891	0.00077	1
EV only	4	1359	0	0.94444	0.00053	0.99891	0.00115	1

This ablation is therefore a sample-size outlook rather than a claim that EV information has no value. At the current calibration scale, EV bins fragment the calibration set: the most detailed EV-conditioned key has only 33 samples in its smallest calibration family, compared with 198 for the primary topology+PV+loading key. If the same scenario design were expanded while keeping the EV family definition fixed, the EV-conditioned split would need roughly six times more calibration support before its weakest family matches the primary grouping. The supplementary sensitivity curve below records this threshold logic and should be used to decide when EV-conditioned calibration is worth retesting.

Table 12: EV family sample-size outlook.

Scale	Min EV calib.	Status
1x	33	Fragmented
2x	66	Diagnostic only
4x	132	Retest candidate
6x	198	Support matched
8x	264	Monitor width/FA

The status labels mean, respectively, that the EV-keyed split is too sparse and wider than the primary split; remains below primary-family support; has enough support to justify retesting; reaches the weakest-family support of the topology+PV+loading key; or should be used only if width and false alarms begin to fall.

6.10. Per-Family Calibration Analysis

Per-family analysis exposes where conditioned calibration works and where it remains weak.

Table 13: Per-family conformal calibration analysis.

Family	Calib. samples	Test samples	Cov.	Wid.	Recall	FA	Missed
33, PV 0.2, high load	198	330	0.97576	0.00069	1.00000	0.00000	0
33, PV 0.4, high load	231	264	0.99621	0.00076	1.00000	0.00000	0

Table 13: Per-family conformal calibration analysis.

Family	Calib. samples	Test samples	Cov.	Wid.	Recall	FA	Missed
33, PV 0.6, high load	396	165	0.84242	0.00070	0.98305	0.01887	1
33, PV 0.8, low load	198	99	0.76768	0.00060	–	0.00000	0
69, PV 0.2, high load	1104	759	0.91173	0.00029	1.00000	0.00000	0
69, PV 0.4, low load	414	483	0.82402	0.00022	1.00000	0.00000	0

The low-coverage 33-bus PV 0.8 low-load family has no actual violating buses, so it is a calibration-efficiency warning rather than a missed-risk warning. The 33-bus PV 0.6 high-load family is the main risk-relevant weak family: it contains 59 violating buses and one missed violation. These rows motivate the discussion: conditioned calibration is useful, but it does not remove the need for sufficient family-level calibration samples and explicit per-family reporting.

We therefore add a post-hoc recalibration audit rather than hiding this weak family. The audit uses test AC labels only for diagnosis and asks how much an inflate-only family radius update would be needed to restore the 90% empirical target. Across 16 topology/PV/loading families, five are below nominal coverage. Inflating only those undercovered family radii increases weighted interval width from 0.000433 to 0.000451 p.u. and leaves false alarms and missed violations unchanged in this representative split. For the weakest coverage family, 33-bus PV 0.8 low-load, coverage improves from 0.76768 to 0.90909 without affecting violations because the family has no violating buses. For the risk-relevant 33-bus PV 0.6 high-load family, coverage improves from 0.84242 to 0.90909 while the single missed violation remains. This is the proper operational reading of conditioned conformal calibration: family reporting identifies where explicit recalibration changes coverage and sharpness, and whether the undercovered family is operationally risky or merely interval-inefficient.

Table 14: Family recalibration audit.

Audit	Families	Under-cov.	Miss cur.	Miss audit	Width cur.	Width audit	FA cur.	FA audit
Inflate undercovered families only	16	5	1	1	0.000433	0.000451	0.00058	0.00058

6.11. Risk-Stratified Calibration and Screening

Aggregate coverage can hide the voltage-boundary behavior that matters for operation. We therefore add a risk-stratified audit on the representative random split. Bus samples and scenarios are

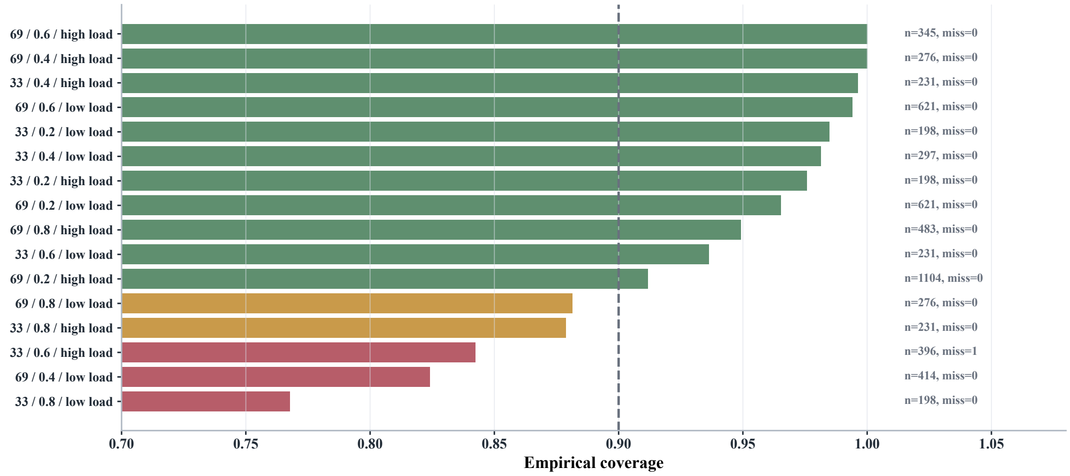

Figure 4: Per-family conditioned calibration audit. The bars show empirical coverage by topology/PV/loading family with sample counts and missed violations, exposing the small-family regimes where shrinkage matters.

grouped into true violations, boundary-safe cases within 0.002 p.u. of either voltage limit, near-safe cases within 0.010 p.u., and interior-safe cases. This audit is derived from saved raw interval predictions and does not retrain any estimator.

Table 15: Risk-stratified calibration audit.

Method	Stratum	Bus samples	Cov.	Wid.	Risk rate	Recall/FA	Missed / FAs
Boost-GC	violating	921	0.59501	0.00110	0.99674	0.99674 recall	3 missed
VoltGuard	violating	921	0.87188	0.00052	0.99891	0.99891 recall	1 missed
GNN ablation	violating	921	0.74159	0.00611	0.99783	0.99783 recall	2 missed
Boost-GC	boundary-safe	55	0.78182	0.00110	0.23636	0.23636 FA	13 FAs
VoltGuard	boundary-safe	55	0.92727	0.00044	0.05455	0.05455 FA	3 FAs
GNN ablation	boundary-safe	55	0.90909	0.00664	0.81818	0.81818 FA	45 FAs

The boundary-safe rows are important for operating value. A conservative interval can achieve high recall by flagging many near-limit but safe buses, which increases unnecessary downstream AC calls or corrective actions. VoltGuard reduces boundary-safe false-alarm buses from 13 under boosting plus global conformal and 45 under the neural graph ablation to three, while still missing only one of 921 truly violating buses. At scenario level, VoltGuard recalls all 93 truly risky scenarios and produces zero false-alarm scenarios across the boundary-safe, near-safe, and interior-safe strata. This is a more direct voltage-risk-screening test than aggregate RMSE alone.

6.12. *Shift and Scenario-Level Results*

Under PV-penetration shift, VoltGuard keeps bus-level violation recall at 99.81%, but empirical coverage drops below nominal because the test distribution is intentionally shifted. This is exactly why the paper does not claim unconditional distribution-free safety. Under topology-held-out transfer from 33-bus training to 69-bus calibration/testing, the topology+PV+loading variant has zero missed bus-level violations across the three seeds and lower false alarms than boosting plus global conformal. This is encouraging, but it is still a controlled 33-to-69 transfer with calibration on the target feeder, not a proof of arbitrary topology-shift safety.

We also audit an explicit PV-shift recalibration protocol. The point estimator is still trained only on the low/medium-PV split. A small, disjoint subset of early high-PV scenarios is then treated as target calibration, and the remaining high-PV scenarios are evaluated. This simulates periodic AC-audited recalibration after a feeder enters a higher renewable-hosting regime; the target calibration labels are not reused for testing.

Table 16: PV-shift recalibration protocol.

PV protocol	Target calib.	Hi-PV test	Cov.	Wid.	Recall	FA	Missed
Src only	0	132	0.76915	0.00081	0.99807	0.00147	1.00
Src+10% target	16	116	0.89527	0.00189	1.00000	0.00398	0.00
Src+20% target	29	103	0.88790	0.00176	1.00000	0.00322	0.00

Thus target recalibration repairs much of the coverage loss and removes the remaining missed bus-level violation, but it does so by widening intervals and raising false alarms. This is not an unconditional conformal guarantee under PV shift; it is evidence that the screening layer can be re-calibrated when a new high-PV operating regime supplies AC-audited calibration cases.

The same PV-shift adaptation is also translated into operating screening quantities on the remaining high-PV scenarios. With source-only calibration, VoltGuard releases 57.00 high-PV scenarios from downstream AC optimization on average, but still has one missed bus-level violation. Adding the 10% target high-PV calibration set removes the bus-level missed violation while keeping scenario-level risky recall at 1.00000. The cost is operationally visible: screened-safe releases fall to 49.67 scenarios and the mean interval width increases from 0.00081 to 0.00209 p.u. This is the desired interpretation for energy management: target recalibration improves high-PV risk conservatism, but it consumes calibration data and reduces the number of scenarios that can be safely bypassed.

Table 17: PV-shift operating-screening protocol.

PV protocol	Target calib.	AC avoided	Risky recall	Post miss	Missed mean	Width mean
Src only	0	57.00	1.00000	0.00000	1.00	0.00081
Src+10% target	16	49.67	1.00000	0.00000	0.00	0.00209

At scenario level, the representative random split has 120 test scenarios and 93 risky scenarios. VoltGuard topology+PV+loading conditioned calibration has scenario recall 1.000, zero missed risky scenarios, and zero scenario false alarms. The neural graph ablation also avoids missed risky scenarios but triggers many more scenario false alarms and has worse bus-level sharpness.

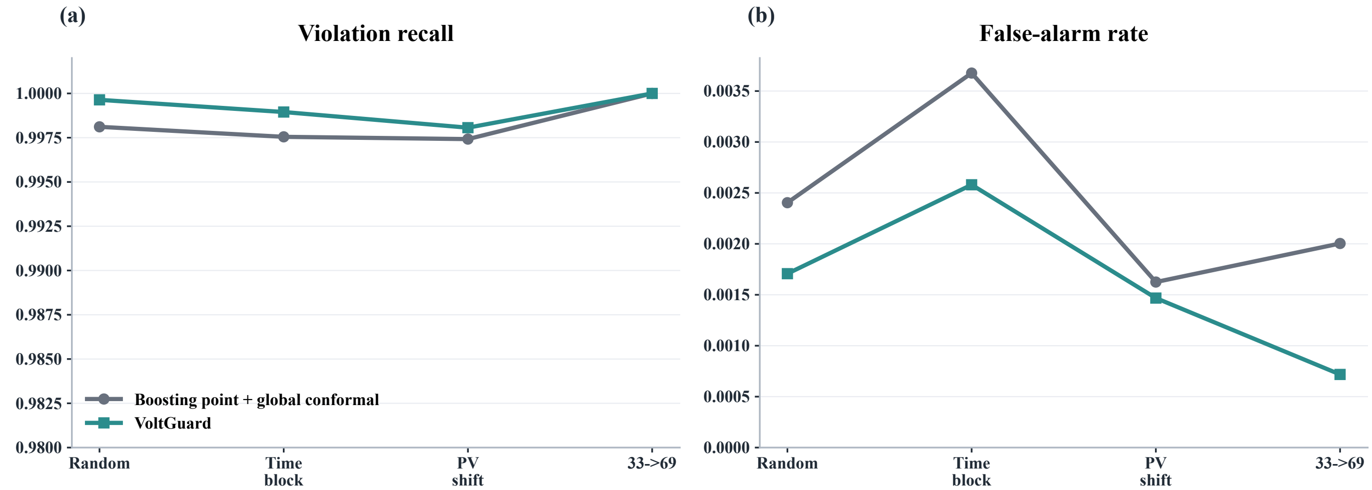

Figure 5: Shift generalization across time, PV penetration, and topology transfer. (a) Multi-seed violation recall under random, synthetic time-block, PV-shift, and topology-transfer splits; (b) corresponding false-alarm rates.

6.13. Expanded IEEE 118-Bus Stress Audit

The IEEE 118-bus case is not used as the main distribution-network evidence, but the review requested either a realistic feeder case or a more complete analysis of this supplementary system. We therefore report the same representative random-split metrics on the 118-bus stress case. The result is intentionally interpreted as a stress audit because the LinDistFlow assumptions and radial distribution-feeder interpretation are not native to this system.

Table 18: IEEE 118-bus supplementary stress audit.

Method	Variant	RMSE	Cov.	Wid.	Recall	FA	Missed
LDF	none	0.88711	n/a	n/a	0.80723	0.87352	48
RF	none	0.00445	n/a	n/a	0.58635	0.00425	103
GB	none	0.00466	n/a	n/a	0.57831	0.00586	105
Boost-GC	Global	0.00466	0.89237	0.01005	0.82731	0.06236	43
VoltGuard	Global	0.00572	0.90254	0.01054	0.81526	0.06851	46
VoltGuard	PV-cond	0.00572	0.91243	0.01211	0.88755	0.07159	28
VoltGuard	T/PV/L	0.00572	0.91328	0.01211	0.88353	0.07100	29

The 118-bus audit does not strengthen the distribution-feeder claim; it exposes where the present physical backbone is inappropriate. The stress case requires much wider intervals and still misses more bus-level violations than the 33/69 distribution-feeder experiments. This is why the main evidence is kept on IEEE 33-bus and 69-bus feeders, while realistic unbalanced OpenDSS/SMART-DS feeders remain a required next validation step rather than a claim made by the current manuscript.

6.14. Failure Mode Analysis

The representative 33/69 random split contains one missed bus-level violation for the selected topology/PV/loading conditioned VoltGuard interval. The miss occurs on IEEE 33-bus scenario 153, bus 7, under PV bin 0.6 and high loading. The true voltage is 0.949940 p.u., only 0.000060 p.u. below the 0.95 p.u. limit. The predicted voltage is 0.950518 p.u.; with a conformal radius of 0.000351 p.u., the lower interval endpoint is 0.950167 p.u. and therefore just misses the undervoltage boundary.

This is not a missed risky scenario. The same scenario contains many deeper undervoltage buses, including bus 17 at 0.916314 p.u. and bus 16 at 0.917032 p.u., all of which are flagged by the interval screen. The failure mode is therefore a boundary-bus miss inside an already risky scenario, not a release of a dangerous scenario. Operationally, the scenario would still enter the corrective-audit queue because other buses are flagged.

The early-warning indicators are visible in the inputs: high loading, PV bin 0.6, multiple downstream buses near the undervoltage band, and a scenario whose minimum interval lower bound is already far below 0.95 p.u. The DMS integration paper turns these indicators into fallback rules: stale telemetry, unseen topology, out-of-envelope forecast features, and rolling residual alarms force watch or corrective-audit routing even when an individual boundary bus is not flagged.

6.15. Computational Cost and Breakeven Analysis

The reviewer-requested timing audit separates training, online screening, and downstream AC-audited grid search. On the representative random split, fitting the residual learner uses 18,360 training rows and takes 1.858 s; the full train/calibration/test evaluation takes 4.468 s. The supplementary 118-bus stress run takes 2.426 s for model fitting and 2.508 s end-to-end.

Table 19: Operational runtime benchmark.

Workflow	AC scen.	Seconds	ms/scenario	Speedup
VG screen	120	0.03046	0.25380	6796.17
Full AC all	120	206.98291	1724.86	1.00
VG flag+AC	93	160.44221	1725.19	1.29
VG top20+AC	24	41.42704	1726.13	5.00

A rough breakeven calculation is favorable because online screening is cheap. Using the 1.858 s fit time and the per-scenario difference between full AC grid search and screening-only evaluation, the fit cost is recovered after roughly 1.1 screened scenarios. The more important operational question is not training amortization but triage policy: conservative intervals may still send many scenarios to AC analysis, while risk-budgeted queues can reduce wall-clock time when the operator has a limited AC-study budget.

7. Discussion

The central result is a corrected claim, not a broader one. The best executed model is not a neural graph residual. It is a physics-informed, topology-aware residual learner with conditioned conformal calibration. This makes the paper more defensible: the method's value comes from aligning the residual features and calibration families with feeder and operating regime, not from claiming that message passing alone solves voltage security.

The conformal guarantee is bounded. Split conformal prediction provides finite-sample coverage under exchangeability of calibration and test samples. Grid data can violate this assumption when topology, load regimes, PV penetration, or time periods shift. The PV-shift and topology-held-out experiments make that limitation visible rather than hiding it. VoltGuard should therefore be used as a calibrated screening estimator under a stated scenario family and recalibration protocol, not as a universal safety certificate.

The screening interpretation is also bounded. The intervals can prioritize scenarios and buses for downstream AC analysis, but they do not certify AC feasibility, replace state estimation, or select final dispatch. OPF, MPC, and AC power flow remain the authority for corrective action. This distinction is important for reviewer interpretation: the paper's claim is sharper intervals and fewer missed voltage-risk events than baseline screening methods, not a new distribution-grid controller.

Model maintenance is part of the operating design. A deployed screen should store every model version, feeder model, calibration family, and release/watch decision, then sample released scenarios for periodic AC audit. Recalibration should be triggered when rolling residuals exceed the calibration-family threshold, when topology identifiers are unseen, when DER capacity or siting changes materially, when reconductoring/regulator/capacitor settings change, or when forecast distributions move outside the training envelope. In those cases the DMS should disable release or fall back to a global conservative radius until a fresh AC-audited calibration set is available.

Real-world validation remains the most important external gap. The present evidence uses public IEEE-style test systems and synthetic operating scenarios; it does not include an anonymized utility feeder, real SCADA/AMI streams, or an OpenDSS/SMART-DS unbalanced feeder with regulator, capacitor, and phase connectivity details. On real feeders, missing telemetry, phase imbalance, equipment controls, and data-governance rules may dominate the residual structure. The current results should therefore be read as a reproducible method demonstration and screening benchmark, while utility-pilot validation is required before operational deployment claims.

The present feeder models are balanced single-phase equivalents, so extension to highly unbalanced three-phase distribution systems is not automatic. On IEEE 123-bus or OpenDSS feeders, the LinDistFlow backbone would need a phase-resolved branch-flow approximation with phase-specific voltage limits, mutual coupling terms, regulator and capacitor states, and missing-phase connectivity. The residual learner can still use the same philosophy, but its features would become phase-indexed and equipment-aware, and the conformal families would likely need to include regulator configuration, phase loading, and feeder section. This is a limitation of the current 33/69 evidence.

8. Conclusion

This paper presents VoltGuard, a physics-informed topology-aware residual learning framework with conformal calibration for voltage-risk screening in active distribution networks. On IEEE 33-bus and IEEE 69-bus feeder scenarios, the topology+PV+loading conditioned conformal variant improves missed-violation and interval-sharpness behavior relative to global conformal calibration.

The work deliberately keeps neural graph residual learning as an ablation because the executed neural model does not justify being the central claim. Future work will extend the same residual-plus-conditioned-calibration framework to unbalanced IEEE 123-bus/OpenDSS or SMART-DS feeders and evaluate recalibration under larger topology and DER regime shifts.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Code and Data Availability

The experiments use publicly available IEEE test systems implemented through pandapower and a project-local IEEE 69-bus feeder implementation. The local reproducibility package includes the scenario generator, configured evaluation pipeline, conformal calibration code, raw predictions, conformal scores, runtime tables, post-action AC audit outputs, DMS prototype logs, reviewer-requested baseline comparisons, and energy-management value metrics. File-level checksums and reproduction commands are provided in two reproducibility manifest files stored under the project results directory. The Python implementation, synthetic scenario-generation scripts, trained model artifacts, table-generation scripts, manuscript sources, and release PDFs are archived in the GitHub project Zhaosiqiang / voltguard-voltage-risk-screening, release v1.0.0-submission, and on Zenodo with DOI 10.5281/zenodo.21149702.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used AI-assisted coding and language tools for drafting support, code scaffolding, consistency checks, and language refinement. The authors reviewed and edited all generated text and code, verified the equations and numerical claims, and take full responsibility for the content of the article.

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